ECON100B - SECTION 15

Introduction to the Short Run Part II & Recent Macro-cylsms

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ANNOUNCEMENTS/AGENDA

Announcements

- Take Home Midterm “Essay” due Friday, 3/20

Agenda

- Intro to the Short Run Part II
- Recent Macro-clysms Part I

Question of the Day

Suppose we have an economy with a Long Run predicted output of 20,000 and an actual output of 22,000. What is the short run output of this economy?

Are you a lake person or ocean person?

The Short Run

The Short Run model

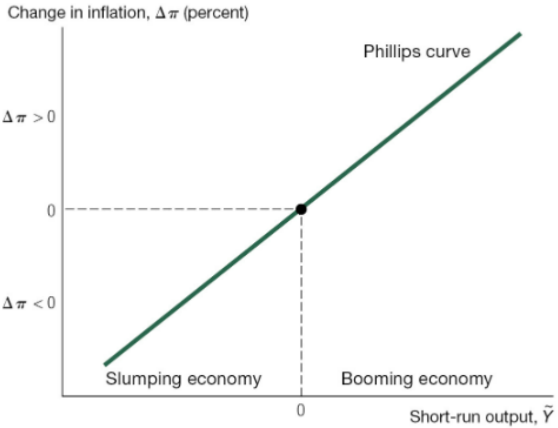
- A model with many actors: consumers and workers, firms, government, and (simplified) trade with the rest of the world

Assumptions of the Short-run model

1. The **economy is constantly being hit by shocks** that push output and/or inflation from potential output or inflation's long-run value
2. **Monetary and fiscal policy affect output**; policymakers have an effect on neutralizing or creating shocks to the economy
3. There is a dynamic **trade-off between output and inflation**: if an economy produces more than its potential output, it leads to an increase in the inflation rate (Phillips Curve)

The Phillips curve

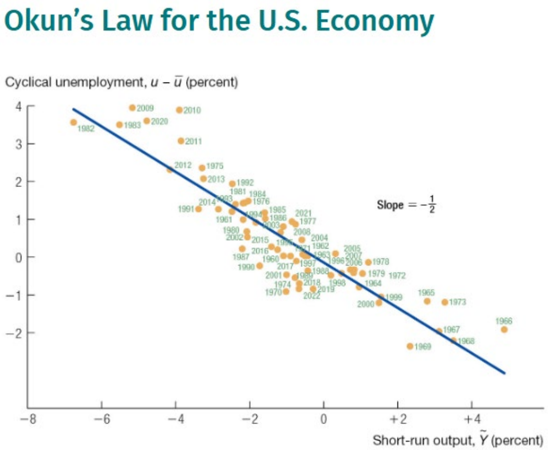
The Phillips Curve



Inflation and Output

- Why is there higher inflation when actual output is more than potential output?
- Potential output reflects how much our output should be under normal economic conditions
- But imagine an economy where the demand for a good is super high: firms might choose to expand their production (at higher costs) and consumers might be willing to pay more; both dynamics end up driving up prices
- Imagine that type of effect across an entire economy, causing inflation to rise
- Empirically, on average, we say that when output exceeds potential by 3 percent, inflation rates rise by about 1 percentage point

Okun's Law



Source: FRED. U.S. Department of Commerce.

Okun's Law

- Generally, when we have higher short-run output, we see a lower rate of cyclical unemployment (which makes sense!)
- We can map the empirical relationship with the following formula:

$$u - \bar{u} = -\frac{1}{2} \times \tilde{Y},$$

- In the rest of the short-run part, we'll focus on output, but you can think about the relationship of output on unemployment as well!

The Short Run Practice

- Suppose the economy is undergoing a period of high inflation. What actions should the Federal Reserve take, and what consequences would this have on the economy? When have we seen this happen before?
- Suppose our economy is slumping, and our governments wants to help improve the economic conditions. What actions can the government take, and what potential consequences would this have on the economy?
- Suppose our short-run output is 4%. What does Okun’s law state will happen to our cyclical unemployment rate?

The Short Run Practice Answers

- They can raise the interest rate to reduce inflation, but the Phillips Curve states that it will probably decrease output as well. This is what happened during the 1970s + 1980s inflation crisis and subsequent Volcker Shock
- Fiscal policy to try to increase output, but could lead to higher inflation by the Phillips Curve
- -1/2 * 4% = -2%. So, we would expect a 2% drop in cyclical unemployment per Okun's Law

Recent Macro-clysms

Recent Shocks

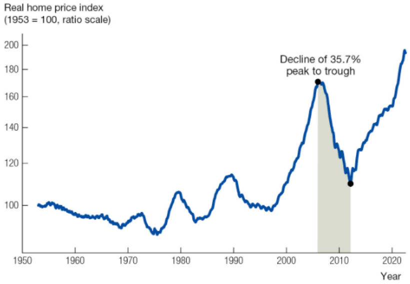
- In the past two decades, there’s been two generational economic shocks that we’ve seen in the United States (and much of the globe)
- **The Great Recession/Global Financial Crisis of 2008-2009**
    - Worst financial crisis since the Great Depression in the 1930s
- **The Covid-19 pandemic that started in 2020**
    - Worst health crisis since the 1918 flu pandemic

The Great Recession

- Banks had made enormous profits in the first half of the 2000s decades, driven by risky mortgages and mortgage-backed securities
- The housing price decline which began in 2006 caused a huge number of defaults on these mortgages
- Many major banks either folded, required government bailouts, or got capital infusions from the government
- Unemployment rose from 4.4% in March 2007 to a peak of 10% in October 2009

Housing Prices

- Housing prices **increased by a factor of nearly 3** between 1996 and 2006, or around 10% per year
- This was fueled by a booming economy, low interest rates, and loose lending standards
- Post-crisis, the national index for housing prices declined by 36% between 2006 – first quarter of 2012



Source: Robert Shiller, econ.yale.edu/~shiller/data/Fig3-1.xls.

The Global Saving Glut

- Financial crises in the 1990s prompted shifts in the macroeconomies of developing countries
- Previously, many of these countries were investing more than they were savings, financed by borrowing from the rest of the world
- Many of these developing countries faced financial crises in the 1990s, causing them to **increase saving substantially** and curtail foreign borrowers, **becoming lenders to places like the U.S.** instead of borrowers
- Ben Bernanke, who was to become the Fed chair during the Recession argued that this caused a **global saving glut**: there was a lot of money that looked to save in good investment opportunities

Subprime Lending

- Many people took out loans for housing during this period (partly driven by global saving glut and low interest rates)
- Some of these borrowers were subprime borrowers, people who had poor credit records or high debt-to-income ratios
- As long as housing prices increased, people could refinance based on the higher equity of the house so they wouldn't default on the loans

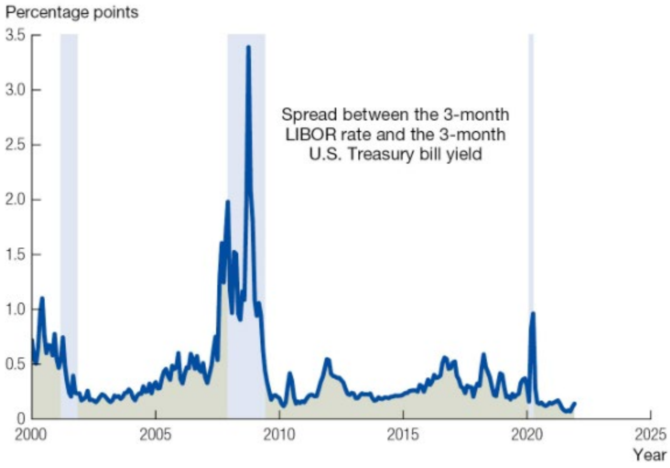
- The federal reserve then voted to increase interest rates in an effort to slow down inflation
    - But this also slowed down the housing market due to higher interest rates
    - Housing prices no longer perpetually increased in price, causing defaults on loans, creating a vicious cycle of lowering housing values

Where Banks Come in

- Banks securitized their assets, lumping financial instruments like mortgages to sell to investors (MBS)
- This process is meant to insulate investors from the risk of individual assets (like one person's mortgage)
- But once the housing bubble burst, many of these mortgages were defaulted on
- A lot of the investors at the end were banks and financial institutions!
- They lost a lot of money on these MBS once the housing market went belly-up, causing ripple effects for other banks, businesses, and individuals
- There was a **liquidity crisis** as banks and other financial institutions became more nervous to lend to each other, businesses, and individuals
    - This is known as charging a **risk premium**

Liquidity Crisis

Liquidity and Risk Shocks from Interest Rate Spreads



The Response

- The government had to take over Fannie Mae and Freddie Mac
- The Federal Reserve organized an $85B bailout of AIG
- Lehman Brothers collapsed into bankruptcy
- Other banks were sold to other banks or received major cash infusions
- The S&P 500 fell more than 50% between 16 months from November 2007 – March 2009

The Great Recession Questions

1. What happened to housing prices between 1996-2006? What happened to housing prices following 2006?
2. What were Mortgage Backed Securities? What were the assumptions behind Mortgage Backed Securities that made them appear safe? How were these assumptions violated during the Great Recession
3. Why did the failure of Mortgage Backed Securities lead to the Financial Crisis?
4. What is a liquidity crisis?

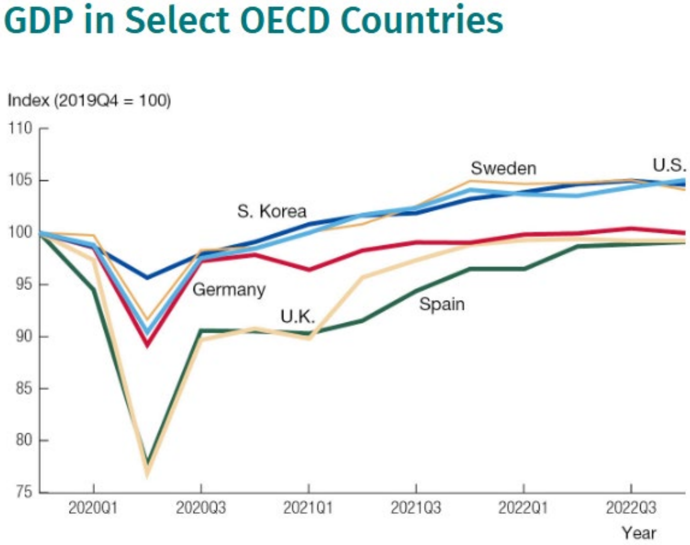
The Great Recession Answers

1. Housing prices between 1996-2006 increased dramatically, and appeared like they would rise in perpetuity, leading to lots of subprime mortgages. When the housing market took a hit, many people defaulted on mortgages, causing the housing market to fall
2. See slides, MBS are securitized collections of mortgages sold to investors. They appeared safe because they aggregated demand, but this was violated in the housing market crash
3. MBS failure led to banks losing a lot of money, causing them to fail to pay back loans, bankruptcy, etc.
4. A liquidity crisis is when the volume of transactions in a financial market falls sharply such that it becomes impossible to borrow money

The Covid-19 Pandemic

- For a middle aged American, the EPA uses (used?) a figure of $10M to value the life of a typical middle-aged American
- The loss of life and expected lifespan from the pandemic contributed to a hit of $750,000 to the lifetime consumption of an average American
- Educational costs may affect the annual consumption of affected cohorts
- In the U.S. there was an estimated loss of 3.7% of a year's GDP, while in some countries this was higher than 10%. On average, this was around 6%

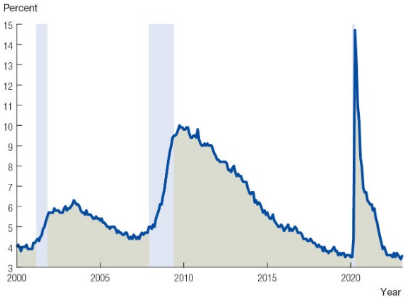
How Covid affected GDP



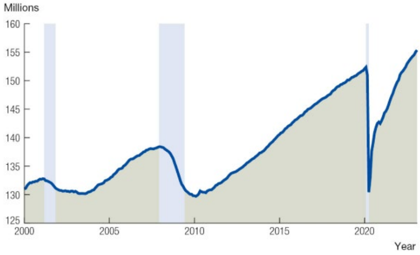
Impact of these Macroeconomic Events

Impact on Employment

The U.S. Unemployment Rate

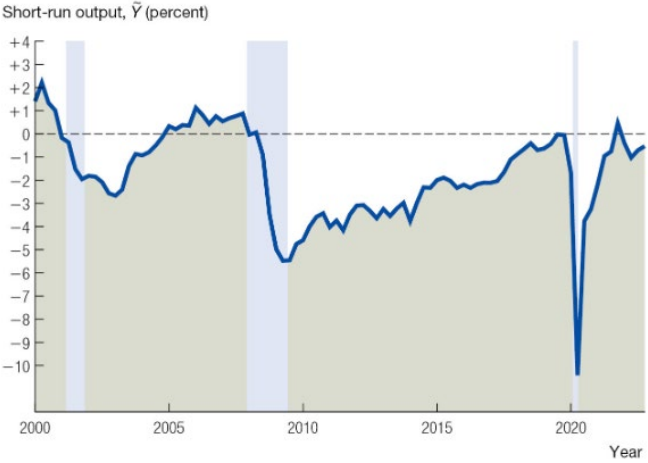


Nonfarm Employment in the U.S. Economy



Short Run Output Impact

U.S. Short-Run Output, $\tilde{Y}$



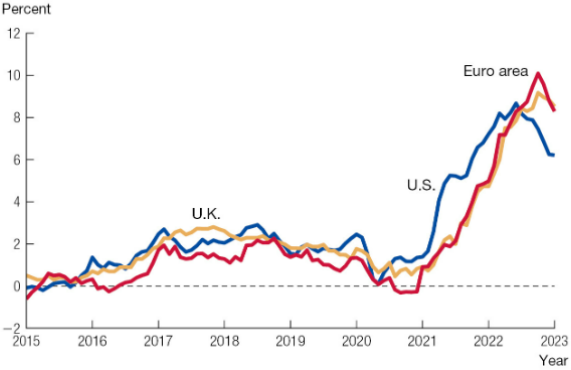
Source: The FRED database and author's calculations.

Impact on Recessions

	Average of previous recessions since 1950	Great Recession	Covid-19 Recession
GDP	-1.7%	-4.7%	-9.6%
Nonfarm employment	-2.5%	-6.3%	-11.9%
Unemployment rate	2.5 pp	4.5 pp	9.4 pp
Components of GDP			
Consumption	0.4%	-3.4%	-10.6%
Investment	-14.4%	-34.0%	-16.5%
Government purchases	1.2%	5.5%	2.6%
Exports	-1.5%	-10.3%	-24.1%
Imports	-4.2%	-18.7%	-20.2%

Inflation (CPI)

Consumer Price Index Inflation in the U.S., U.K., and Euro Area



Source: Federal Reserve Economic Data (FRED). Inflation is calculated using monthly data as the percent change over the preceding 12 months.

Intro to Financial Economics

Balance Sheet

TABLE 10.3

A Hypothetical Bank's Balance Sheet (billions of dollars)

Assets		Liabilities	
Loans	1,000	Deposits	1,000
Investments	900	Short-term debt	400
Cash and reserves	100	Long-term debt	400
Total assets	2,000	*Total liabilities*	1,800
		Equity (net worth)	200

Bank Regulations

- A reserve requirement mandates that banks keep a certain percentage of deposits on reserve with the central bank (to make sure they have money to pay out in the case of a run)
- Capital requirements mandate that the capital/net worth of the bank be at least a certain fraction of the bank's total assets (For example reserves and cash)

Leverage

- Leverage refers to the ratio of total liabilities to net worth
- Leverage magnifies change in value of assets and liabilities in terms of returns to shareholders

- This can play out in mortgages/home ownership as well
- Imagine a house that costs $500,000 where a homeowner puts down a 20% downpayment, such that they start with $100,000 equity in the home
- If the price of the house rises by 10%, the equity is now worth $150,000, such that the homeowner has 50% more equity

- For major investment banks, the leverage ratios were incredibly high, such that a small aggregate shock could make them insolvent
- Systemic risk, as banks are all tied into each other